

Job market signaling.

Spence (1973), Spence (1974), Hörner (2008), Mas-Colell et al. (1995).

Outline of the game

- Nature determines whether an agent has low or high productivity.
- The agent privately observes his/her productivity and choose a level of education. (Player 1)
- The “labor market” (i.e. player 2) observes the education chosen by the worker and offers a wage.
- Worker accept or reject the offer.

Spence's job market signaling

- Nature chooses the agent's productivity $\eta \in \{\eta_\ell, \eta_h\}$, $0 < \eta_\ell < \eta_h$; with probability $\lambda \in (0, 1)$, $\eta = \eta_h$.
- After privately observing η , the agent chooses $e \in \{0, 1\}$.
Education, that is $e = 1$, costs c_ℓ to type η_ℓ and c_h to type η_h ; no education ($e = 0$) costs zero to both types.
- After observing e , *the market or the employer* offers a wage $w \in \mathbb{R}$ and the agent accepts or rejects the offer.

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- A worker that receives w and pays c has a payoff $w - c$.

If not employed, the agent's payoff is $0 - c$.

- An employer who hires a worker with productivity η and pays w has payoff $\eta - w$.

If no worker is hired, the employer's payoff is zero. Employers are risk neutral.

A signaling game

Player 2

Various alternatives to model “the market.”

- Reduced form. The market wage will satisfy a zero expected profit condition where expectation is taken with respect to the probability generated by λ, σ_1 .
- Explicit. After observing e , two (or more) firms simultaneously and independently offer a wage. Bertrand competition then insures a zero expected profit condition in any sequential equilibrium.
- Fictitious player 2 with payoff $U_2(\eta, e, w) = -(\eta - w)^2$

Benchmark

- No information. No player observes η .

Equilibrium: $(e = 0, w = (1 - \lambda)\eta_\ell + \lambda\eta - h)$.

- Complete information. Both players observe η .

Equilibrium: $(e = 0, w = \eta)$

Efficiency ex ante and efficiency ex post.

Separating equilibrium

Proposed equilibrium.

- Player 1. $(s_1(\eta_l), s_1(\eta_h)) = (0, 1)$
- Player 2. $(s_2(0), s_2(1)) = (w_0, w_1) = (\eta_l, \eta_h)$

s_2 is sequentially rational.

- Player 2's information sets $e = 0$ and $e = 1$ are reached with probability $(1 - \lambda) > 0, \lambda > 0$ respectively.
- Player 2's beliefs are $\mu(\eta_l|e = 0) = 1, \mu(\eta_h|e = 1) = 1$. Bayes rule.
- Zero profit implies $w_0 = \eta_l, w_1 = \eta_h$.

It remains to show that s_1 is sequentially rational

Incentive compatibility

It remains to show that s_1 is sequentially rational

- player 1, type η_ℓ .

$$U_1(\eta_\ell, e = 0, s_2(0)) \geq U_1(\eta_\ell, e = 1, s_2(1))$$
$$\eta_\ell \geq \eta_h - c_\ell$$

Player 1, η_ℓ *incentives are compatible* with proposed strategies if the inequality is satisfied.

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- player 1, type η_h .

$$U_1(\eta_h, e = 1, s_2(1)) \geq U_1(\eta_h, e = 0, s_2(0))$$
$$\eta_h - c_h \geq \eta_\ell$$

Player 1, η_h with proposed strategies if the inequality is satisfied.

The proposed strategies are a separating equilibrium if

$$\eta_h - c_h \geq \eta_\ell \geq \eta_h - c_\ell$$

This implies $c_h \leq c_\ell$.

Pooling equilibrium

Proposed equilibrium

- Player 1. $(s_1(\eta_\ell), s_1(\eta_h)) = (1, 1)$
- Player 2. $(s_2(0), s_2(1)) = (w_0, w_1) = (\eta_\ell, \bar{\eta}_b)$ where

$$\bar{\eta}_b = (1 - \lambda)\eta_\ell + \lambda\eta_h.$$

s_2 is sequentially rational.

- Player 2's information sets $e = 0$ is not on the path of play, $e = 1$ is reached with probability 1.
- Player 2's beliefs when $e = 1$ are $\mu(\eta_\ell|e = 1) = 1 - \lambda$, $\mu(\eta_h|e = 1) = \lambda$. Bayes rule.
- Player 2's beliefs when $e = 0$ are $\mu(\eta_\ell|e = 0) = 1$.
- Zero expected profit condition implies $w_0 = \eta_\ell$, $w_1 = \bar{\eta}_b$.

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Pooling equilibrium

It remains to show that s_1 is sequentially rational.

Player 1, type η_ℓ :

$$U_1(\eta_\ell, e = 1, s_2(1)) \geq U_1(\eta_\ell, e = 0, s_2(0))$$
$$\bar{\eta}_b - c_\ell \geq \eta_\ell$$

Player 1, type η_h :

$$U_1(\eta_h, e = 1, s_2(1)) \geq U_1(\eta_h, e = 0, s_2(0))$$
$$\bar{\eta}_b - c_h \geq \eta_\ell$$

Remarks

- Separating equilibrium is suboptimal. Education is not an argument in payoff functions.
- P.O. allocation is pooling equilibrium without education.
- At P.O. allocation high-productivity types subsidize low productivity types.

References

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